10 The theory theory

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This is a book about domain-specific cognition – the proposal that at least some human conceptual abilities are specialized for some types of contents and not for others. In this chapter we address the development of a single domain: everyday understanding of the mind. We suggest that this development is best understood as the formulation of a succession of naive theories. Moreover, this "theory theory" can help to characterize cognitive domains more generally and to explain domain-specific development. Our chapter, therefore, joins company with a number of recent discussions drawing parallels between theory change in science and cognitive development (Carey, 1985, 1988; Gopnik, 1984, 1988; Karmiloff-Smith & Inhelder, 1974; Keil, 1987; Wellman, 1985, 1990).

Theory of mind

In the past five years there has been an explosion of interest in children's early understanding of the mind (Butterworth, Harris, Leslie & Wellman, 1991; Astington, Harris, & Olson, 1988; Frye & Moore, 1991; Whiten, 1991). The research tackling this question has come to be called "children's theory of mind." This title reflects a leading explanatory position among investigators in this area. According to this position, our everyday conception of the mind is an implicit naive theory; children's early conceptions of the mind are also implicit theories, and changes in those conceptions are theory changes. We refer to this explanatory position as the "theory theory."

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If the theory theory has any content, it should be falsifiable. It should lead to empirical predictions about the course of development, and these predictions should be different from those of alternative accounts. We will describe both the nature and predictions of the theory theory in some detail and contrast them with two alternative accounts, a "simulation theory" and an "innate module theory." Some of the predictions of the theory theory are similar to the predictions of one or the other of these alternatives. However, the overall pattern of predictions is distinctive. In the end we suggest that the current data, by and large, support the theory theory.

What is a theory?

When we say that children's understanding of the mind is a theory, we do not necessarily mean that it is an explicitly formulated theory, or that the cognitive enterprises children engage in are identical to those of adult scientists. The question of when and how children come to do science is an interesting one (Kuhn, 1989), but it is not the one we consider here.[1] There are obvious differences between scientists and children. For example, scientific change, as it's usually conceived, takes place over a period of many years and involves the entire scientific community, whereas cognitive change takes place in individual children in a much shorter time period.

Nevertheless, scientific theory change and conceptual change in childhood are both the product of human minds trying to understand the world around them. Scientific change must centrally involve some human cognitive capacity. But although this claim seems nearly self-evident to psychologists, it has not been the prevailing view in philosophy of science itself. Instead, philosophers of science have typically been concerned with logical questions about the nature of scientific knowledge, or, more recently, with sociological and historical questions about the nature of change in the scientific community. To oversimplify, we might say that the original project of classical philosophy of science was to try to show some logical way in which true theories could be constructed from evidence, in the way, for example, that complex logical systems could be constructed from simple axioms. The failure of this project led to a view that emphasized the largely arbitrary, socially determined nature of actual scientific change.

A cognitive interpretation of science might hold some middle ground between these views. On this view scientific theory change may be neither logically necessary nor sociologically arbitrary. Instead, it reflects cognitive processes that genuinely lead to a veridical conception of the world, though not necessarily in a way that is logically guaranteed. Curiously, this view has really only begun to be elaborated in the philosophy of science itself. The cognitive science of science is still in its infancy. Nevertheless, the large body of phenomena described in literature on the history and philosophy of science

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will inform any such cognitivist view, and any such view will have to provide an account of those phenomena.

From this cognitive point of view, both cognitive developmentalists and philosophers of science face a similar question – how to characterize mechanisms for learning about the nature of the world. Two of the clearest and most impressive instances we have of such learning are the acquisitions of children and the achievements of science. How do children and scientists develop basic knowledge about the world? The hypothesis of the theory theory is that there are deep similarities between the underlying cognitive mechanisms involved in the epistemological endeavors of childhood and of science. Philosophers of science are developing a characterization of cognitive structures – scientific theories, paradigms, research programs – that organize, produce, and store scientific knowledge. Developmental psychologists are developing a parallel notion of everyday theories and research programs that organize, produce, and store everyday knowledge.

Insights from philosophy of science, then, can help to illuminate the epistemological achievements of childhood. But the similarities are better captured by thinking of scientists as big children, rather than thinking of children as little scientists. The progress of science, we believe, reflects certain fundamental processes of conceptual change that are first seen in very young children.

Within philosophy of science there is much argument over what theories are and how they change. In spite of these controversies, there are some very general features that, on almost all conceptions, are characteristic of theories and distinguish them from other types of knowledge. Our reference list will include strange bedfellows to a philosopher of science, including such classic accounts as those of Nagel and Popper as well as revisionist views such as those of Laudan and Kuhn. This reflects the tension between logical and sociological views of science.

Nevertheless both schools have provided us with rich descriptions of the actual course of theory formation and change in science. In what follows we highlight the features of theories and theory change in science that seem to us common across these very different accounts and that seem most relevant and helpful in understanding cognitive development. We do not aim for a definitive account of scientific theories. On the contrary, the point of the theory theory is not to demonstrate that what scientists do is identical with what children do, but to use the scientific notions as a touchstone for characterizing an important sort of cognitive development.

Empirical typologies versus theories: Abstractness and coherence

Theories are always constructed with reference to evidence, some layer of experience different from the theories themselves (Nagel, 1961; Lakatos, 1970; Laudan, 1977; Popper, 1965). The relation between theory

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and evidence is distinctive. It is not like the relation between experience and other cognitive structures. To capture this, we distinguish between two ways of organizing experience. The first kind we will call "empirical typologies and generalizations," the second "theories." Empirical typologies and generalizations are orderings, partitionings, and glosses of evidence and experience, but they share the same basic vocabulary as the evidence itself. For example, I may look at a collection of plants and divide them into ones with no stems, ones with green stems, and ones with woody stems. Similarly, I may make generalizations about the behavior of objects. If the evening star appears at the same place in the heavens every evening we may make the empirical generalization that it will continue to appear at that place this evening (without any theoretical notions of why this may be the case). If on many occasions moldy bread relieves infected wounds, we might make the generalization that mold relieves infection (Nagel, 1961; Hempel, 1965). In developmental psychology the idea that children have "scripts" or "schemas" (e.g., Nelson, 1985) implies that much early cognition consists of empirical typologies and generalizations. And this may well be true for certain sorts of knowledge.

Empirical typologies and generalizations contrast with theoretical structures. Theories propose theoretical constructs: Abstract entities, events, or forces. These theoretical entities do explanatory work – they provide causal explanations that account for evidential phenomena. These theoretical explanations are typically phrased in a vocabulary that is quite different from the evidential vocabulary.

For example, suppose we postulate Darwinian species as a theoretical construct in evolutionary biology (rather than describing birds, mammals, etc., as empirical types). We will define species in terms that are quite removed from their apparent features; a green stemmed plant and a woody stemmed one may both be ferns because of their reproductive lineage. Similarly, Kepler's theory of the planets postulates elliptical movements that are notoriously not visible when we look at the stars' perceived motions in the sky. In theoretical medicine, we postulate unseen entities, like viruses and bacteria, with distinctive properties. Theoretical constructs are often called unobservable but they need not be definitively unobservable; rather they are appeals to a set of entities removed from, and underlying, the evidential phenomena themselves. Theoretical constructs are designed to explain (not merely type and generalize) those empirical phenomena. So, one characteristic of theories is their abstractness.[2] Theories postulate abstract entities and laws that explain the data but are phrased in a different vocabulary than the data themselves.

Moreover, theoretical constructs do not work independently – they work together in systems characterized by lawlike structures. A second characteristic of theories, therefore, which distinguishes them from empirical generalizations, is their coherence (Hempel, 1965; Quine & Ullian, 1970). The theoretical entities and terms postulated by a theory are closely, "lawfully," interrelated

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with one another. As a result, changes in one part of a theory have consequences for other parts of the theory.

Explanation. The coherence and abstractness of theories together give them an explanatory force that empirical typologies and generalizations lack. Although empirical typologies and generalizations may be useful, they lack deep explanatory power. To "explain" that this plant is stemless because it belongs in the category of plants without stems, or that the star appeared there by saying that it always appeared there before, is legitimate but shallow. In contrast, explaining the position of the evening star in terms of Kepler's theory, or the properties of plants in terms of their evolutionary history, is, at least, cognitively satisfying in a way that empirical generalizations and typologies are not. Moreover, there are reasons to believe that such explanations are also more accurate. They get us closer to the truth.

Prediction. Similarly, empirical generalizations allow predictions, but the predictions are not far removed from the evidence itself. Moreover, they are phrased (roughly) in the same terms as the evidence itself. If the evening star was the first to appear in the heavens in the past, then it will be the first to appear there tonight (Nagel, 1961). In contrast, the abstractness and coherence of theories give them a very characteristic sort of predictiveness. A theory, in contrast to an empirical generalization, a script, or a schema, makes predictions about a wide variety of evidence, including evidence that played no role in the theory's initial construction. In evolutionary biology, knowing a plant's species enables us to predict what its offspring will look like. Kepler's account allows one to predict the behavior of new celestial objects, moons, for example, which were quite unknown at the time the theory was formulated. Theories in medicine allow us to predict that antibiotics will inhibit many bacterial infections, including some, like scarlet fever, that present none of the symptoms of an infected wound, or some, like Legionnaire's disease, that were unknown when the theory was formulated. They also allow us to predict that such drugs will be useless against viral infections, even when the symptoms of the viral infection are identical to those of a bacterial one.

Some of these predictions will be correct. The theory will accurately predict future events described at the evidential level, and will do so in ways that no single empirical generalization could capture. Others will be incorrect. Because theories go beyond the evidence, and because theories are never completely right, some of their predictions will be falsified. In still other cases the theory will make no prediction at all.

In fact, the theory may in some circumstances have less predictive power than a very large set of empirical generalizations would. This is because explanatory depth and force do not simply equate with predictive accuracy. We can make predictions about events without explaining them: Kepler's theory still leaves many of Tycho Brahe's observations unexplained; there are no theoretical explanations for the efficacy of aspirin, and yet empirical

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generalizations about its effects are the basis for an industry. Theoretical predictions are different in two ways. First, a few theoretical entities and laws can lead to a wide variety of unexpected predictions. Second, in the case of a theory, prediction is intimately tied to explanation.

Interpretation. As we've seen empirical typologies and generalizations are closely related to the evidence itself. A taxonomy or generalization may call our attention to certain features of the evidence rather than others, but it will rarely lead us to actively misconstrue the evidence. In contrast, theories produce interpretations of evidence, not simply descriptions and typologies of evidence and generalizations about it. It is notoriously true that theoretical preconceptions may lead a scientist to dismiss some kinds of evidence as simply noise, or to reinterpret others as suspect or the result of methodological failures (Lakatos, 1970; Scheffler, 1967). Nevertheless, these influences of the theory on evidential interpretation are not merely arbitrary. There are characteristic ways in which theory and evidence interact (Glymour, 1980).

All of these characteristics of theories ought also to apply to children's understanding of mind, if such understandings are theories of mind. That is, children's theories should involve appeal to abstract unobservable entities, with coherent relations among them. Theories should invoke characteristic explanations phrased in terms of these abstract entities and laws. They should also lead to characteristic patterns of predictions, including predictions about new types of evidence and false predictions. Finally, theories should lead to distinctive interpretations of evidence; a child with one theory should interpret even fundamental facts and experiences differently than a child with a different theory. This distinctive pattern of explanation, prediction, and interpretation is among the best indicators of a theoretical structure, and it provides a way of distinguishing the theory theory from some of its competitors.

Dynamic features of theories. So far we have been talking mostly about the static features of theories, the features that can distinguish them from other cognitive structures such as typologies or schemas. We turn now to theory changes. A most important thing about theories is what philosophers call their defeasibility. Theories are open to defeat or revision via evidence and because of this theories change. In fact, a tenet of modern epistemology is that any aspect of a theory, even the most central ones, may change (Quine, 1961; Laudan, 1977).

Theories change as a result of a number of different epistemological processes. One particularly critical factor is the accumulation of counterevidence to the theory (Popper, 1965). The mere fact of counterevidence, however, may not lead to the theory's rejection; instead the interpretive mechanisms of the theory may simply reinterpret the counterevidence (Lakatos, 1970). This fact has sometimes led to a kind of epistemological nihilism, as if theory change was just a matter of caprice, or sociology (Feyerabend, 1975). But

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even though a precise logical specification of theory change may elude us, there are certainly substantive things to be said about how it typically takes place. There are characteristic intermediate processes involved in the transition from one theory to another (Kuhn, 1962; Lakatos, 1970; Laudan, 1977).

The initial reaction, as it were, of a theory to counterevidence may be a kind of denial. The interpretive mechanisms of the theory may treat the counterevidence as noise, as something not worth attending to. However, at a slightly later stage the theory may call on ad hoc auxiliary hypotheses designed to account specifically for such counterevidence. Typically, these hypotheses are only invoked to explain particular cases of counterevidence and are not applied more generally. Auxiliary hypotheses may also be helpful because they phrase the counterevidence in the accepted vocabulary of the earlier theory. Such auxiliary hypotheses, however, often appear, over time, to undermine the coherence that is one of a theory's strengths. The theory gets ugly and gerrymandered instead of being elegant and tight.

A third step requires the availability or formulation of some alternative model to the original theory. A theory may limp along for some time under the weight of its auxiliary hypotheses if no alternative way of making progress is available. Even when a viable alternative emerges, its fertility is typically not recognized immediately. The first inklings of the new conceptual structure may appear in the form of relatively small modifications of the conceptual apparatus of the earlier theory. Only later may it become clear that the new idea also provides an alternative explanation for the evidence that was central to the earlier theory.

The development of the heliocentric theory of planetary movement provides some good examples of these intermediate processes. Auxiliary hypotheses involving more and more complex arrangements of epicycles were invoked to deal with specific pieces of counterevidence. At the same time, a new conception of planetary movement, heliocentrism, was beginning to emerge. It is worth noting though that Copernicus's theory itself failed to apply the heliocentric idea very widely. In many respects Copernicus's account is more like the Ptolemaic one, than, say, Tycho Brahe's account. It includes epicycles, for example. Brahe's account acknowledges many of the flaws of the Ptolemaic ones, and uses the idea of heliocentrism to deal with them (other planets revolve around the sun, which revolves around the earth). But even Brahe fails to accept the central idea that the earth itself goes around the sun. Only with Kepler is there a really coherent heliocentric account that deals both with the anomalies and with the earlier data itself.

We propose that these same dynamic features should be apparent in children's transition from one theory to a later one, and specifically from one view of the mind to another. Children should ignore certain kinds of counterevidence initially. Later, they might account for them by auxiliary hypotheses, only invoked to deal with particular kinds of counterevidence. Furthermore, children might first use the new theoretical idea in limited contexts, contexts

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closely related to the conceptual structure of the earlier theory. Only after the theory change is complete will they finally reorganize their knowledge so that new theoretical entities play a central role.

The child's theories of mind

In this section we briefly outline an emerging picture of children's early understanding of the mind. The description we provide is of a succession of theories, the child's developing theories of mind. The description helps to clarify the theory theory, by outlining one sensible version of the theory for further discussion. The data also fit such a characterization, we believe; one of the strengths of the theory theory is that it usefully organizes a welter of emerging findings.

A number of misconceptions about children's theories of mind are beginning to appear. One is that there is a single "child's theory of mind" that emerges at some point in development. A second is that possessing this theory of mind should be identified with solving the Wimmer and Perner (1983) false-belief task. A third is that before developing this theory of mind the child is a behaviorist. We share none of these assumptions. In fact, we believe that they are inconsistent with the theory theory and the data. As we have seen an essential feature of theories is that they change. In theory change one can identify a succession of different, progressively more accurate theories. Moreover, on the theory theory, no single piece of knowledge, or performance on a single task, should be critical. The important thing about a theory is that it allows inferences about a wide variety of problems. Nor is there any reason to suppose that children are ever behaviorists – the theories we will discuss are all mentalistic ones.

We propose that there is a change from one mentalistic psychological theory to another somewhere between $2\frac{1}{2}$ and to 4 years of age.[3] The change is not a simple all or none one, but rather involves a more gradual transition from one view of the mind to another. Indeed this change manifests the telltale intermediate processes that are characteristic of theory change. Two-year-olds have an early theory including a conception of mental states such as desires and perceptions, but this theory is incorrect from an adult perspective, in that it does not posit the existence of representational mental states, prototypically beliefs. In 3-year-olds we see an intermediate phase. Children develop a nonrepresentational account of belief. They also begin to understand representational aspects of their well-developed notions of perception and desire. And they begin to countenance mental representations as a kind of auxiliary hypothesis, invoked to explain salient counterevidence. In a third phase, beginning around the age of 4, children reorganize their central explanatory theory. They begin to realize that what actors think – their representation of the world rather than the world itself – inevitably determines their actions.

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The 2-year-old theory

The 2-year-old is clearly a mentalist and not a behaviorist (e.g., Wellman & Woolley, 1990). Indeed, it seems unlikely to us that there is ever a time when normal children are behaviorists. Even in infancy, children seem to have some notions, however vague, of internal psychological states. This is evidenced in very early "conversational" interaction and in facial imitation; almost from birth infants coordinate their actions with those of others in a sensitive and fine-grained way (Meltzoff & Gopnik, 1993; Gopnik, 1993). Later, more clearly, infant mentalism is demonstrated in social referencing and joint attention behaviors; 9-month-olds seek information about other's psychological relations to objects (Wellman, 1993). It seems plausible, therefore, that mentalism is the starting state of psychological knowledge. However, this early mentalistic theory does not include all the sorts of mental states that adults understand.

More specifically, even at 2 years, psychological knowledge seems to be structured largely in terms of two types of internal states, desires, on the one hand, and perceptions, on the other. Desire and perception alone provide examples of the two basic categories of explanatory entities in folk psychology – the two types of theoretical constructs that Searle (1983) calls "world-to-mind" and "mind-to-world" states. An understanding of desire encompasses an early knowledge that what is in the mind can change what is in the world. An understanding of perception, on the other hand, encompasses an early knowledge that what is in the mind depends on what is in the world. Moreover, both desire and perception, as theoretical constructs, work to explain action but may also be divorced from any particular actions that an agent may perform.

More important, however, desire and perception can be, and at first are, understood in non-representational terms. Desires at first are conceived simply as something like drives toward objects (Wellman & Woolley, 1990). Perceptions are at first understood simply as awareness of or visual contact with objects (Flavell, 1988). In neither case need the child conceive of a complex propositional or representational relationship between these mental states and the world. Instead, these very young children seem to treat desire and perception as simple causal links between the mind and the world. Given that an agent desires an object, the agent will act to obtain it. Given that an object is within a viewer's line of sight, the viewer will see it.

These causal constructs are simple, but they have considerable predictive power. In particular, together they allow the first form of "the practical syllogism": "If an agent desires x, and sees that x exists, he will do things to get x." Even that form of the practical syllogism is a powerful inferential folk psychological law. It allows children to infer, for example, that if John wants a cookie and sees one in the cookie jar, he will go there for it. If he doesn't want it, or doesn't see it, he won't. And there is evidence that $2^1/_2$-year-olds make just such inferences (Wellman, 1990, chap. 8).

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The 3-year-old theory

By age 3, children begin to show signs of a more elaborate mental ontology. Natural language can provide us with one avenue for exploring these abilities. Before 3, children make extensive and appropriate use of terms for desire and perception (Bretherton & Beeghly, 1982). More cognitive mental terms (think, know, remember, make-believe, dream) only begin to emerge around the third birthday (Shatz, Wellman, & Silber, 1983). There is further evidence that at age 3 children begin to develop a more general notion of belief (Moses, 1993; Wellman & Bartsch, 1988), and also representational but fictional mental states such as pretenses, dreams, and images (Wellman & Estes, 1986; Woolley & Wellman, 1992, 1993).

When these concepts first appear, however, they have an interesting character, framed by the child's larger theory that is still a desire-perception theory. Initially the understanding of belief is largely nonrepresentational, and is modeled on an earlier understanding of desire and especially perception. On this view, belief, like perception and desire, involves rather direct causal links between objects and believers. This view has variously been called a "copy theory" (Wellman, 1990), a "Gibsonian theory" (Astington & Gopnik, 1991b), a "situation theory" (Perner, 1991), or a "cognitive connection" (Flavell, 1988) theory of belief. The similar idea in all these accounts is that belief contents directly reflect the world. In these respects the child's first conception of belief seems to be a relatively simple extension of earlier nonrepresentational theoretical constructs.

The most well-known evidence that children have this nonrepresentational view of belief is their misunderstanding of situations that involve misrepresentation. In the classic false-belief task, for example, children may see a candy box that has pencils instead of candy inside. They are asked what someone else will think is inside the box. Three-year-olds typically say "pencils." They behave as if there is a simple and reliable causal link between the real state of affairs in the world, and our mental states about it. As we will see 3-year-olds also behave as if they hold this view in a variety of superficially quite different tasks. So the first understanding of belief is arguably nonrepresentational in a crucial fashion. The 3-year-old's conception of fictional mental states such as imaginings and dreams suffers from a similar restriction. Three-year-olds (but not 4-year-olds) tend to expect even fictional imagings to reflect the real world truthfully (Woolley & Wellman, 1993).

Conversely, as the third year progresses, children begin to understand representational aspects of desire and perception. Desire and perception may be construed either nonrepresentationally, or representationally. In fact, in the adult theory, desire and perception are as representational as belief. What we want and see (by and large) is not the thing itself but the thing as represented. Understanding some aspects of desire and perception seems to require

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this sort of representational understanding. For example when we are satiated with something we no longer desire it, but the object itself has not changed; when different types of people have different tastes or values, their desires differ but the objects of desire remain the same; when people see the same object right-side up or upside-down, or through different colored filters, their perceptions of the object may differ. As we will see there is evidence that these representational aspects of desire and perception may be understood earlier than equivalently representational aspects of belief.

The 5-year-old-view

By age 4 or 5, children, at least in our culture, have developed a quite different view of the mind, one that we and others have called a "representational model of mind" (Flavell, 1988; Forguson & Gopnik, 1988; Perner, 1991). On this view almost all psychological functioning is mediated by representations. Desires, perceptions, beliefs, pretences, and images all involve a fundamentally similar structure, a structure sometimes described in terms of propositional attitudes and propositional contents. These mental states all involve representations of reality, rather than realities themselves. In philosophical terms, the child's view of the mind becomes fully "intentional." To use Dretske's (1969) terminology, perceiving becomes perceiving that, desiring becomes desiring that, and we would add, believing becomes believing that. This new view provides a kind of Copernican, or better Keplerian, revolution in the child's view of the mind. In addition to distinguishing different types of mental states with different relations to a real world of objects (desires vs. perceptions), the child sees that all mental life has the same representational character. Many characteristics of all mental states can be explained by the properties of representations. For example, representations change over time, and differ among people with different experiences. This newly unified view not only provides new predictions, explanations, and interpretations, but it also eventually provides a revised view of the very phenomena that were accounted for earlier by the desire-perception theory.

The child's theory as theory

What evidence do we have for these claims about the child's conception of the mind? In particular what evidence do we have for thinking that these understandings are theoretical in the sense that we have been outlining so far? We will argue that (1) the child's understanding involves general constructs about the mind that go beyond any direct evidence. (2) These constructs feature prominently in explanation. (3) The constructs allow children to make predictions about behavior in a wide variety of circumstances, including predictions about behavior they have never actually experienced and incorrect predictions. (4) Finally, these constructs lead to distinctive interpretations of evidence.

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Explanations

Children's explanations of action show a characteristic theorylike pattern. In open-ended explanation tasks (Bartsch & Wellman, 1989; Wellman & Banerjee, 1991) children are simply presented an action or reaction ("Jane is looking for her kitty under the piano") and asked to explain it ("Why is she doing that?"). There are many mental states that might be associated with such situations. Yet 3- and 4-year-old children's answers to such open-ended questions are organized around beliefs and desires just as adults' are ("she wants the kitty"; "she thinks its under the piano"). Moreover, there is a shift in explanatory type between 2 and 5. Two-year-old's explanations frequently mention desires, never beliefs. Asked why the girl looks for her doll under the bed, they will talk about the fact that she wants the doll, but not about the fact that she believes the doll is there. Three-year-olds invoke beliefs and desires, and some 3- and 4- and 5-year-olds consistently refer to the representational character of these states, explaining failure in terms of falsity. These same trends can be seen in the explanations children give in their spontaneous speech (Bartsch & Wellman, 1990, and in press).

Predictions

Consider the desire-perception theory. Even that early theory allows children to make a variety of predictions about actions and perceptions, both their own and others'. For example, they should be able to predict that desires may differ, and that, given a desire, an actor will try to fulfill that desire. They should understand that desires may exist without being acted on, and that they may be unfulfilled. They should predict that fulfilled desires will lead to happiness, whereas unfulfilled desires will lead to sadness. And there is evidence that, in fact, all these kinds of predictions are made by very young children (e.g., Astington & Gopnik, 1991a; Wellman & Woolley, 1990; Yuill, 1984). Similarly, a child with the desire-perception theory should be able to predict the perceptions of others in a wide variety of circumstances, including those in which the perceptions are different from their own. Such very early activities as shared attention (e.g., Bakeman & Adamson, 1984) and social referencing (e.g., Hornik, Risenhoover, & Gunnar, 1987) may already indicate some capacity to predict the perceptions of others. Other aspects of this understanding quickly develop. By $2^1/_2$ these Level-1 predictions, as Flavell calls them, are firmly and reliably in place (Flavell, 1988). At this age children can reliably predict when an agent will or will not see (and hear and touch) an object. They can also predict how seeing an object will lead to later actions.

These predictions may seem so transparent to adults that we think of them not as predictions at all but simply as empirical facts. A little reflection, however, should make us realize that the notion of desire or perception used by these very young children is theoretically broad and powerful. Children

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can use the notion of desire appropriately and make the correct predictions when the desired targets are objects, or events, or states of affairs; they can attribute desires to themselves and others, even when they do not act to fulfill the desires and when the desires are not in fact fulfilled (Astington & Gopnik, 1991a; Wellman & Woolley, 1990). Similarly, children seem to make accurate predictions about perception across a wide range of events, and involving factors as different as screens, blindfolds, and visual angles, and do so across different perceptual modalities (Flavell, 1988). Again, they may do so even when the perceptions do not lead to any immediate observable actions. Moreover, given novel and unfamiliar information about an agent's desires and perceptions, children will make quite accurate predictions about the agent's actions (Astington & Gopnik, 1991a; Wellman & Bartsch, 1988; Wellman & Woolley, 1990).

More significantly, however, these children also make incorrect predictions in cases where the desire-perception theory and its successor, the nonrepresentational theory of belief, breaks down. The most well-known instance of such an incorrect prediction is the false-belief error in 3-year-olds, such as on the candy-pencils task. The focus on false-belief tasks, however, may be unfortunate because it has promoted a mind-set in which any ability to perform "correctly" on a false-belief task is taken as evidence that the child has a representational theory of the mind. As we will see, there are cases in which 3-year-olds indicate some understanding of false belief without having yet formulated a representational theory of the more general sort that we earlier attributed to 4- and 5-year-olds. To begin with, however, it is worth pointing out the much greater ubiquity and generality of incorrect false-belief predictions. Three-year-olds make erroneous predictions, not only in the "classic" tasks, such as the candy-pencils task, but also in many other cases involving beliefs about location, identity, number, and properties. They make incorrect predictions for other children, for adults, for puppets, and for hypothetical story characters. Incorrect predictions are made when the question is phrased in terms of what the other thinks, what the other will say, and what the other will do, and across a wide range of syntactic frames. They are made by American, British, and Austrian children, and by Baka children of the Cameroons (Avis & Harris, 1991; Gopnik & Astington, 1988; Moses & Flavell, 1990; Perner, Leekam, & Wimmer, 1987).

Moreover, and more significantly from the point of view of the theory theory, these incorrect belief predictions are mirrored by similar incorrect predictions on a wide range of other tasks. A brief inventory would begin with appearance-reality tasks. If children are shown a misleading object, say a "hollywood rock" that is really made of sponge, they typically say both that the object really is a sponge and that it looks like a sponge; they fail to distinguish appearance, as represented, from reality. These tasks have themselves proved robust across many variations of culture, question, and material (Flavell, Green, & Flavell, 1986; Flavell, Zhang, Zou, & Qi, 1983).

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Also included are judgments about the sources of beliefs. For example, 3-year-old children fail to predict that someone with limited information about an object will have limited knowledge of the object. O'Neill, Astington, and Flavell (1992) showed children two objects that felt identical but looked different; 3-year-olds had difficulty predicting that someone who saw the objects would be able to differentiate between them, but not someone who only felt them. Young children have similar difficulty understanding subjective probability (Moore, Pure, & Furrow, 1990). Three-year-olds understand that some people may know things that others do not but they fail to appreciate that there may be different degrees of belief. That is, they do not predict any difference in accuracy between someone who is certain of something and someone who merely thinks it may be true. Finally, young children evidence related errors in the understanding of pictorial representation systems. Just as 3-year-olds typically predict that all beliefs must be accurate, they also predict that pictures and photographs must be accurate (Zaitchek, 1990).

Arguably all these tasks require a representational model of the mind. The early nonrepresentational theory of belief leads to incorrect predictions in cases of misrepresentation such as false beliefs, and also false photographs or misleading appearances (consider the similar difficulties for Gibsonian accounts in psychology). But the early theory also, in a quite different way, makes it difficult to understand that beliefs may come from many different sources, that they may comé in degrees, and that there are intermediate steps between the mind and the world.

We might phrase this by saying that young 3-year-olds believe that cognitive states like belief come in only two varieties, total knowledge, when the world is causally related to the mind, and absolute ignorance, when it is not. In cases of misrepresentation, there is one object in the world and two people are both related to that object, but differences in the causal relations between the minds and the world lead to different representations of that object. In the sources and subjective probability cases there is one object in the world, and two people arrive at the same representation of that object, but the causal relations between each person and the world are different. To distinguish between degrees of belief or sources of belief we need to understand that people's cognitive relations to the world may differ in significant ways even when both their ultimate beliefs and the objects in the world are the same. Moreover, the notion of complex intermediate representational processes relating world and mind provides a causal framework for understanding many cases of misrepresentation. (See Astington & Gopnik, 1991b; Gopnik, 1993; Wellman, 1990, for further discussion).

In some of these tasks the nonrepresentational theory of belief makes no predictions and children answer at random. Significantly, however, in many cases the theory makes consistent incorrect predictions, and young children consistently give the same wrong answer, with correct answers emerging at about 4 years. The tasks involve very different types of situations and questions

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and their information-processing demands are quite varied. Moreover, the standard methodology of these studies has included control tasks, involving similar or identical situations, questions, and information-processing demands, which children seem entirely capable of answering. It seems then that children's difficulties are not due to some memory, attention, or language comprehension abilities that are unavailable to younger but available to older subjects. Nor do any dimensions of familiarity, at least in any simple terms, seem to underlie the difference between tasks at which children succeed and tasks at which they fail. Instead all these tasks seem to call on the same theoretical conception of the mind. Insofar as that theoretical conception fails, so do the children.

Interpretations

In the tasks we have described so far children are clearly using mental states to explain and predict, both correctly and incorrectly – two of the central functions of theoretical constructs. Children's responses also show strong interpretive effects. Suppose we present the young child with situations or events that amount to counterevidence for the desire-perception theory? If the child is simply reporting her empirical experience or observations, we might expect that she will report that evidence correctly. In fact, however, children consistently misreport and misinterpret evidence when it conflicts with their theoretical preconceptions. Flavell and his colleagues have some provocative but simple demonstrations of evidential misinterpretation (Flavell, Flavell, Green, & Moses, 1990). A child sees a blue cup, agrees that it is blue and not white, and sees the cup hidden behind a screen. At this point another adult comes into the room, and she says, "I cannot see the cup. Hmm, I think it is white." Then the child is asked what color he thinks the cup is and what color the adult thinks it is. To be correct the child need only report the adult's actual words, but 3-year-olds err by attributing to the adult a true belief. Even if corrected, "well actually she really thinks it's white," they continue to insist the adult has a factually correct belief: "She thinks its blue."

Moreover, in several studies, 3-year-olds make standard false-belief errors in reporting their own false beliefs, errors equivalent to their assessment of others' beliefs. In these cases children need not predict a person's belief or actions, they need only report their own past experienced beliefs. In fact, they misreport, that is misinterpret, their own beliefs. Gopnik and Astington (1988), for example, showed children a variety of deceptive objects, such as a candy box full of pencils. The traditional false-belief question was "What will Nicky (an ignorant other) think is inside the box?" The self-report false-belief question was "When you first saw the box, before we opened it, what did you think was inside?" Three-year-olds erred when reporting their own past false belief just as they did in assessing the other's false belief, and in each case they made the same error, reporting the true belief instead. Similarly,

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3-year-olds are inaccurate in reporting the sources of their own beliefs, just as they fail to predict accurately the effects of sources on the beliefs of others (Gopnik & Graf, 1988; O'Neill & Gopnik, 1991). However, 3-year-olds are very accurate in reporting simple past perceptions (Gopnik & Slaughter, 1991). In short, many 3-year-olds interpret beliefs in terms of a restricted nonrepresentational theory. When their experiences fit the theory, they report them accurately. When their experiences do not fit the theory, they misconstrue and misreport them to make them fit (Gopnik, 1993).

Transitional phenomena

Three-year-olds, however, unlike 2-year-olds, are no longer completely ignorant about representation and belief. Instead, recent evidence suggests that during the period from 3 to 4 many children are in a state of transition between two theories, similar, say, to the 50 years between the publication of *De Revolutionibus* and Kepler's discovery of elliptical orbits.

In our discussion of interpretation we described how children with the earlier theory begin by simply denying the existence of counterevidence: Johnny and I really did think and act as if there were pencils in the candy box when we first saw it. We have also described children's initial nonrepresentational account of belief. This nonrepresentational account of belief extends their original desire-perception psychology and leads to their errors in false-belief tasks. But when and how, we now ask, do the first signs of an understanding of misrepresentation, the centerpiece of the 5-year-old theory, begin to appear? Recall that in the scientific case the crucial idea of the new theory may first be used in order to deal with only particularly salient types of counterevidence, or be used in ways that involve only slight extensions of the earlier theory. There is evidence for these sorts of transitional phenomena in children's understanding of mind in the period from 3 to 4.

For example, it is possible that 3-year-old children achieve a partial notion of mental representations in the case of "non-real" mental states, such as pretences, dreams, and images. Certainly there is evidence that children actually have such fictional mental states as young as 18 months (Leslie, 1987). Evidence that they understand such states, however, becomes clear only later. But, by the third birthday children have some conceptual knowledge of these aspects of mental life, and moreover, they may distinguish such imaginary or hypothetical states from the states of desire and perception (Flavell, Flavell, & Green, 1987; Wellman & Estes, 1986). However, these states appear to play little role in 3-year-olds' (and for that matter adults') explanation of ordinary behavior. Significantly, these states have little causal connection to objects (that, in fact, is what is distinctive about them). Although 3-year-old children see desires as states that modify the world, and perceptions as states that are modified by the world, pretenses, images, and dreams, on their view, bear no causal relation to the world at all. So, advances in

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understanding representational mental states may first appear for mental states that play only a peripheral role in children's causal theory of mind and behavior. Furthermore, there is recent evidence that 3-year-olds' conception of fictional mental states is at first marked by a nonrepresentational conception similar to their first nonrepresentational conception of beliefs. As mentioned earlier, 3-year-olds (but not 4-year-olds) tend to expect even fictional imagings to reflect the real world truthfully (Woolley & Wellman, 1993).

Similarly, it seems that first understandings of genuine misrepresentation are developed as only mild extensions of the concepts of desire and perception central to the earlier theory. Desire and perception may be construed either nonrepresentationally, or representationally. In fact, in the adult theory, desire and perception are as representational as belief. What we want and see (by and large) is not the thing itself but the thing as represented. Understanding some aspects of desire and perception seems to require this sort of representational understanding. For example, when we are satiated with something we no longer desire it, but the object itself has not changed; when different types of people have different tastes or values, their desires differ but the objects of desire remain the same. There is evidence that representational aspects of desire may be understood earlier than equivalently representational aspects of belief. Three-year-olds are significantly better at recognizing and reporting changed desires for the very same object than changed beliefs about the same state of affairs (Gopnik & Slaughter, 1991). Similarly, young children are better at recognizing that two people might have similar desires or preferences for the same object than that they have two different beliefs about the same object (Flavell et al., 1990). However, 3-year-old children still do not perform as well on these tasks as they do on simple nonrepresentational desire tasks (Wellman & Woolley, 1990). Similarly, although nonrepresentational aspects of perception are understood by $2^1/_2$ years, representational ones, such as understanding that an object may look right-side up to one observer and upside-down to another, are only understood later (Flavell, 1988; Gopnik & Slaughter, 1991). However, there is evidence that these aspects of perception are understood before corresponding aspects of belief. Both in Flavell's earlier studies and in our recent studies (Gopnik & Slaughter, 1991; Slaughter & Gopnik, 1992), children were better at misrepresentation tasks involving perception than they were at similar false-belief tasks and appearance/reality tasks.

We have suggested that for the 2-year-old the central theoretical constructs are nonrepresentational desires and perceptions whereas for the 4-year-old they are representational beliefs. Three-year-old transitional understandings seem to include both nonrepresentational accounts of belief and representational accounts of imagination, desire and perception. This is reminiscent of the way that Copernicus and Tycho Brahe mix epicycles and heliocentrism.

There is also evidence that early signs of an understanding of misrepresentation may come only when children are forced to consider salient

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counterevidence to their theory. At times, for example, at least as the fourth year progresses, 3-year-olds are able to recognize the existence of beliefs that clearly misrepresent (e.g., Moses, 1993; Robinson & Mitchell, 1992; Siegal & Beattie, 1991; Wellman & Banerjee, 1991). In particular, Bartsch and Wellman (1989) found, as others had, that 3-year-old children continued to make incorrect false-belief predictions about a person's future actions, predicting persons would act as if they had correct knowledge instead. However, if 3-year-olds were confronted with a person who had already committed an incorrect mistaken action, and were asked to explain this anomalous behavior, at least some of them began to talk about misrepresentation. Mistaken actions constitute anomalies, or counterevidence, to nonrepresentational desire-perception explanations of action. Making this sort of counterevidence particularly salient seemed to help to induce initial representational notions, in this transitional age group. Similarly, in a recent study of children's conception of their own past beliefs, Mitchell and Lacohee (1991) required children to select an explicit physical token of their original belief (e.g., a picture of candy in the candy box) and then were shown the real contents of the box (pencils). When asked about their earlier belief in this experiment, children were confronted with their earlier choice of the physical token, and many were then able to avoid the typical misinterpretation of their earlier belief. That is, these children were able to assemble a beginning sense of misrepresentation if confronted with strong, clear counterevidence to their usually incorrect theoretical interpretation of their past belief.

However, to reiterate, these same 3-year-olds do not often think in terms of misrepresentation or false beliefs. When predicting action they typically, consistently, resistantly act as if the actor's desire along with the objective facts determine action, ignoring the role of false belief. Similarly, when asked the contents of a person's belief, they consistently, resistantly cite the objective facts rather than the subjective representations. The more advanced representational notion of belief initially functions like an auxiliary hypothesis rather than a central theoretical construct. It serves to explain strong counterevidence, under pressure, but is not used generatively to provide new predictions. (Compare the way epicycles function in Ptolemaic and early Copernican theory.) It serves to characterize peripheral phenomena, such as fictional imaginings, before it centrally characterizes beliefs.

Do 3-year-olds really understand false belief then? Did Copernicus really understand planetary movement? The answer in both cases is that the question is a bad one. One of the strengths of the theory theory is that it makes such questions otiose. "Understanding" false belief, or developing an idea of representation, involves the development of a coherent, widely applicable theory. It may be possible to have some elements of that theory, or to apply them in some cases, without operating with the full predictive power of the theory, particularly in a transitional state.

We argue therefore, that the transition from $2^1/_2$ to 5 shows all the signs of

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being a theory change. Although initially the theory protects itself from counterevidence, the force of such counterevidence eventually begins to push the theory in the direction of change. The first signs of the theory shift may emerge when counterevidence is made particularly salient. Moreover, the central concepts of the new theory, representation and belief, may first appear as small modifications of concepts that are already well entrenched in the earlier theory, such as desire and perception. Finally, by age 4 or 5 the new theory has more completely taken over from the old. The predictions are widely and readily applicable to a range of cases.

Alternative accounts

So far we have tried to present a picture of how a theory theory would handle the evidence about children's developing theories of mind. We wish to contend not only that the theory theory can account for the developmental data but also that it does so more adequately than other accounts. In this section we consider specifically two alternative accounts of children's understanding of mind: simulation theory and an innate module account. For each of these alternatives we provide a brief description, concentrating on the sorts of evidence that might distinguish that account from the theory theory. No single type of evidence speaks against these accounts; there is no critical experiment that confirms one theory and disconfirms the others. Still, the accounts do have different empirical implications and there is evidence that helps decide between each alternative and the theory theory.

Empirical understanding of the mind: The simulation theory

One alternative to the theory theory is that the child's understanding of the mind is much more closely linked to evidence than to theory. On this view much of the child's understanding of the mind is more like empirical typologies and generalizations than like theories. How might this look? Consider a commonsense account of a person's report of his or her own states of mind. In this account, the person directly experiences his on-going mental states and simply describes them accordingly – I'm wanting this, believing that, imagining the other, and feeling sad about something else altogether. These descriptions type the mental states as one sort or another. Such reports, it could be argued, serve to describe mental states in an appropriate evidential vocabulary – in this case the language of introspective mental experience – just as describing a fern as green, or woody, or big describes it in an appropriate evidential vocabulary – in that case the language of external perceptual experience. In this sort of description, it could be argued, we do not interpret our own mental states in terms of some theory; we merely experience them and are able to type together certain states as experientially similar.

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All this has a certain plausibility for knowledge of one's own states, but what about our knowledge of others? On this view knowledge of one's own mind is direct and evidential, but knowledge of other minds is indirect and predictive. Still, the sort of prediction required is only that provided by any empirical generalization. According to this view we straightforwardly generalize mental states from our own experience to others. I know what I experience when "I want an apple" and I generalize that state to others, inferring that she also "wants an apple" in analogous circumstances or when she displays analogous behavior.

This empirical account of knowledge of the mind is congenial with a classic philosophical view of the understanding of the mind. This view, originally identified with Descartes but widely accepted since, is that we have some form of privileged "perceptual" or introspective access to our own mental states. Something like this view is also implicit in the extensive "perspective-taking" literature in the Piagetian tradition – the developing child must generalize his or her own experiences to other minds.

Recently, several philosophers (Gordon, 1986; Goldman, 1989) and psychologists (Harris, 1991; Johnson, 1988) have proposed more specific versions of this general position – a proposal they term simulation theory. According to simulation accounts, we use our empirical generalizations and typologies to attribute to others the states we would have ourselves if we were in the other's situation, or were producing their behavior. We simulate their mental states from our own, hence the name of this account. In a bit more detail, the simulation process works as follows. We feed perceptual inputs appropriate to the other's situation (e.g., what he can see or touch or hear) to our own mind. By running our mind through this contrived, imaginary situation we simply experience the mental state that would result in that situation. Then we attribute that state to the other person. We simulate the other person's mental state by running a working model of a mind – our own mind – in a simulated situation and seeing what the model yields.

The general empirical generalization account and the more specific simulation accounts have a number of telling empirical consequences; we will focus on only two here (see Gopnik, 1993; Gopnik & Wellman, 1992, for further discussion). The first concerns the experience of your own mind. On this view, in a very important sense, access to your own mental states is supposed to be direct and empirical, requiring no inference or interpretation, no conceptual intermediaries, no theorizing; you simply experience your state and describe it. A consequence of this view is that one cannot erroneously misinterpret, or misconceive, one's own mental state. One simply reports the experience as being of one evidential type or another.

On the theory theory, in contrast, erroneous self-interpretations are not only possible, they are to be expected. One typical characteristic of theories, after all, is that they allow and even force interpretation of the evidence. If the theoretical prediction and evidence are in conflict it is often the evidence rather than the theory that is reinterpreted, as we noted earlier.

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A second empirical consequence, related to this first one, concerns the course of development. For both theory theory and simulation theory we can predict that there will be development: Initial "easy" problems will be solved correctly before later "hard" ones. But the notions of what will be easy and hard differ dramatically between these two accounts. For simulation theory the critical difference is between states that are directly reported versus those that must be simulated – states of the self versus states of others. A child's errors, again in the tradition of perspective-taking views in development, should occur when they must infer others' states, and can be expected to be essentially "egocentric" errors. That is, the child's early errors consist of not correctly adjusting the child's simulation to the other person's condition, an adjustment that is especially necessary, for example, when self and other are in such different situations that they should be experiencing conflicting mental states.

Note that on a simulation account there is no reason to expect that different mental states should be easier or harder to attribute to others. Take beliefs and desires. Both beliefs and desires are equally available to the child as states of her own mind. At a young age, from a simulation perspective, reading off one's own beliefs and one's own desires should be equally easy; attributing beliefs and attributing desires to someone else should be equally problematic, and equally subject to egocentric error when the other person's beliefs and desires conflict with one's own.

In contrast, for the theory theory the critical difference concerns states that are easy or difficult to conceive of. Earlier we described what we take to be a succession in the child's conceptions of mental states, as the child develops and replaces a succession of theories. Especially important is a difference between an early nonrepresentational understanding of mental states and a later more representational understanding. Early on children have a relatively adequate understanding of nonrepresentational desire-perception states. Later they develop an understanding of the representational state of belief, specifically, and a representational understanding of mind more generally (including a representational understanding of certain aspects of perception and desire). Theoretical conceptions of the sort we have described are equally applicable to the self and others. If a theory has formulated a particular theoretical construct, such as the concept of false representations, it should in principle be able to apply this concept equally well to the child's own behavior and the behavior of others. If the theory does not include this construct, it will not be applicable to either the self or others.

We want to describe several empirical findings that are relevant to the different developmental predictions of the two theories. The first set of findings concern children's ability to understand and report their own mental states. For example, as we noted before, children not only fail to understand that others' beliefs can misrepresent; they also fail to understand that their own beliefs can misrepresent. Children who say that the other person will think there are pencils in the candy box, also say that they themselves thought

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there were pencils in the candy box when they first saw it. Children who are unable to understand the sources of others' beliefs are also unable to report the sources of their own beliefs. Children seem to misinterpret even their own experience when it conflicts with their theories.

One crucial comparison concerns children's understanding of desires and perceptions versus belief. In three different tasks Gopnik and Slaughter (1991) presented children with situations in which their desires were satiated and so changed. For example, initially the child desired one of two short books. That one was read to him and the child said he now desired the other book. The test question for the past desires was just like the one for past beliefs: "When you first saw the books, before we read one, which one did you want?" In these tasks 3-year-old children were considerably better at reporting past now-changed desires than past now-changed beliefs. Similarly, Gopnik and Slaughter presented children with situations in which their perception was changed. Children saw an object on one side of a screen and they were then moved to the other side of the screen where they saw a different object. They asked, "When you first sat on the chair, before we moved over here, what did you see on the table?" The children were completely able to report their past perception. Again these same children misreported their own past beliefs.

These experiments concern children's reports of their own mental states, beliefs, desires, and perceptions. From a simulation point of view, why do the children make errors when they are, supposedly, simply reading off their own mental states? And why do they make errors for one state but not for the others?

Perhaps the trouble is that the questions require not a report of current mental states, but a (faulty) memory of past states. Harris (1991) has suggested this. However, the crucial data here concern the comparison between children's reports of belief and reports of other states such as desire and perception. These data suggest that young children can report mental states that are just past but now different, for example, they do so for changed desires and perceptions. The poor performance for beliefs that are just past but now different, therefore, cannot just be a memory problem. This finding presents a paradox for simulation accounts. Either reporting immediately past mental states involves directly reporting one's own states or it involves simulating immediately past experiences. If reporting these immediately past states requires simulation, and simulation is hard for 3-year-olds, then why are 3-year-olds so good at simulating their past desires and perceptions? If reporting past states does not require simulation, because these states are just read off, then why are the 3-year-olds so bad at reading off past beliefs?

More generally, the simulation account suggests that children are at first generally egocentric about the mind and then overcome this by learning that they must adjust their simulations for others. The developmental data do not fit this general mold; there is evidence for nonegocentric understanding quite early for some states. We have already described one such set of tasks, the

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early "Level-1" perspective-taking tasks, in which children can predict that the other child will not see what they see themselves. Similarly, quite young children can predict that someone else will have a desire different from their own (Wellman & Woolley, 1990). One issue for simulation theory, then, must be to explain why children, who can obviously "adjust their simulations" for some states, do not do so for beliefs. Indeed, even for belief itself, the data do not suggest that children's main difficulty involves misattributing their own beliefs to others. Instead, it involves a failure to understand that beliefs can misrepresent.

In this regard a task that makes a particularly interesting contrast to the false-belief task is a "discrepant belief" task (Wellman & Bartsch, 1988). Imagine a task in which the child sees objects at two different locations: There are pencils in the drawer and there are also pencils in the cupboard. The child is told about another person who has seen only the target objects at one location (or alternately that the other person believes there are pencils in only one location). Where will that person look if he wants to find the object? Compare that task to an equivalent false-belief task. In this case the target objects are only in one location, A. The child is told or shown that the other person believes (falsely) that the targets are in location B instead. Where will the other look if he wants to find the target objects? Three-year-olds typically answer correctly on discrepant belief tasks while failing false-belief tasks. On discrepant belief tasks they do not predict that the other would do what they actually do on the basis of their own knowledge (look in both places or in either place indifferently). Instead they predict, correctly, that the other will act on his own belief, one that is discrepant from the child's own belief. At the same time these same children fail the false-belief tasks, asserting that the other will look where the object really is.

In the discrepant belief task the other person possesses a belief that is different from the subject's, but that belief does not misrepresent. There are, in fact, pencils in the location as the other person believes. In this case 3-year-olds pass. In the comparable false-belief task the other's belief is not only different from the child's own belief, it misrepresents; it falsely claims there are candies in the box when there are none. Children's performance on these matched tasks strongly suggests that the problem is not that the child fails to adjust to the other's situation, but it is that the child fails to realize that beliefs can misrepresent.

This is only one example of results that suggest that young children's errors in understanding the mind are not really "egocentric." Even very young children are quite able to attribute to others mental states different from their own. Instead, they err by sometimes misunderstanding what certain mental states are really like.

A related empirical problem is that the simulation theory has difficulty explaining the structure of the explanations that children offer. In children's natural language and particularly their explanations for aberrant actions, they

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initially explain actions only in terms of desires and perceptions. As they grow older belief explanations also begin to emerge, occasionally including false-belief explanations (Bartsch & Wellman, 1989; in press). From a simulation point of view, the child's own mind, even at the very youngest ages, is a device that itself contains states like beliefs as well as desires. The child's model outputs both beliefs and desires. Why should children's explanations and predictions first privilege desires over beliefs? There is no reason to expect this if the child is simply running simulations and reporting outcomes. On the theory theory, in contrast, there is a good reason why children's explanations and predictions at first ignore beliefs and especially false beliefs or misrepresentations. Young children have yet to come to a theoretical conception of belief as an explanatory psychological construct.

To be clear, our preference for the theory theory over simulation accounts does not mean that we completely reject a role for first-person knowledge in the formation and application of a theory of mind. Our own first-person psychological experience provides a body of data that strongly informs and shapes our concepts of mind. Our claim is that these first-person experiences are also theory-laden. First-person knowledge of our own mental states is not merely experiential, it is a product of conceptual interpretation intertwined with self-experience.

Similarly, we do not wish to deny an important role for the simulation of others' states of mind. A theory of mind must confront the problem of inferring the mental states of others in particular cases. Several devices can be used to aid in this process, but imagining what your own mental states would be rates high among them. We contend, however, that such simulations are theory-driven. To understand another's actions we simulate his or her beliefs and desires. Why those mental states among the myriad possibilities? Why not simulate his or her daydreams, night dreams, aches, and sensations? Our simulations, we argue, are information acquisition attempts framed by a theory. Theories direct their holders to look for some sorts of data and not others; our theory of mind directs us to look for or construct some sorts of information about others. Simulations in this view are a consequence of our theory of the mind.

The innate module theory

A second serious alternative to the theory theory is that an understanding of mind is the result of an innate module. In this theory, basic mental concepts such as belief and desire are not constructed from evidence in the course of childhood development. Instead, an innate structure dedicated to interpreting behavior in terms of beliefs and desires has been forged phylogenetically, in evolution. Ontogenetically, that structure may need to be triggered by certain sorts of inputs, but once triggered it creates mandatory interpretations of human behavior in mentalistic terms. Something like this

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alternative can be derived from Fodor's writings about modules (1983) and his position that belief-desire conceptions of the mind are innate to humans (1987, 1992). Moreover, Leslie (1987, 1991) proposes that there is a distinct theory of mind module that makes its first developmental appearance with the onset of pretence in 18-month-olds.

There are two separable aspects of this alternative, as we see it. The first is the claim that there is innate knowledge about the mind. The second is the claim that this knowledge is modular. We believe that there is innate knowledge about the mind but we do not believe there is a theory of mind module (see also Astington & Gopnik, 1991; Gopnik, 1993; Meltzoff & Gopnik, 1993).

A naive theory may begin from an innate starting state, an initial mapping of inputs onto representations that is innately specified. However, from the theory theory point of view any starting conceptions would themselves be revised and reorganized as a result of countervailing evidence. What is innate is an initial state, a first theory. We are born with certain kinds of psychological knowledge that begin a process of theory development and revision. Moreover, if children do not have these initial representations, this first theory (in the case of autism, for example) we might expect them to develop a further theory of the mind in radically different ways, if they develop such a theory at all. Such a view seems implicit in the claims of some investigators of theory of mind in autism, such as Baron-Cohen (1990) and Hobson (1989). On this view, however, such initial structures, although innate, would be defeasible; any part of them could be modified or altered with new evidence. It makes sense to us to talk about innate theories, and to see the process of theory change and replacement as one that begins at birth.

When Leslie and Fodor argue for a theory of mind module, however, they are making a much stronger claim. In Fodor's (1983) analysis, for example, modules are not only initially innately specified, but their processing is mandatory, encapsulated, perceptual, and very fast. In addition, modular representations, unlike central knowledge and belief processes, are insensitive to revision via evidence. Along these lines, Leslie, for example (1991), argues that there is an innately specified privileged system of representations of psychological knowledge, analogous to the proposed specialized representational systems of the visual (Marr, 1982) or syntactic (Chomsky, 1975, 1980) systems. This theory of mind module automatically maps given perceptual inputs (say a person's behavior) onto a more abstract set of representations (representations of the actor's mental states). It automatically mandates certain "inferences" or outputs and not others.

It is the mandatory, nonrevisable character of modular processing that we wish to focus on for now. These features mean that the basic representations of the module are, in philosophical terminology, indefeasible. Given a normal brain, once the module has matured and has been triggered, certain representations of the mind will result. Conversely, other representations simply could not be formulated, no matter how much evidence supported them.

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The crucial differences between this account and the theory theory, therefore, concern the interplay of experience and conceptual structure. In theories, experience can radically alter the nature of the theoretical concepts. Evidence about planetary movements can lead to the transformation of a geocentric conception into a heliocentric one. In modules the relation between experience and conceptual representation is different – experience is simply represented in this fashion. Relevant experience can trigger our use of a privileged representational system (or not), but the evidential nature of the experience does not reshape and reconstruct such privileged representations themselves.[4]

We would resist calling indefeasible, modular, cognitive structures theories. However, the representations these structures produce may mimic theoretical inferences or representations in various ways. In particular, like theories, modules invoke abstract entities and rules that are only related to perceptual experience in indirect ways. Also like theories, modules allow predictions that go beyond experience, and allow, indeed mandate, the subject to represent evidence in a particular way, a process that may look like interpretation. In modularity accounts, however, the inputs to the representations and the representations themselves, are not related to one another because of their logical or inferential or evidential properties, but simply as a matter of brute biological fact. Modular representations do not lead to predictions because of some set of inductive or deductive generalizations, but because they are designed by evolution to do so.

What kinds of evidence could differentiate between the modularity account and the theory theory? It is worth pointing out that many kinds of evidence that are commonly adduced to support one position or another, in fact, cannot discriminate between these two. The mere fact that there is some psychological knowledge in very early infancy is compatible with either view – that is, with either an innate starting-state theory or an early innate module. In particular, the theory of mind deficits in a biological disorder like autism might stem either from the absence of a theory of mind module, or from the absence of an initial starting-state theory of persons. The fact that the representations in question are abstract, and removed from the evidence of actual experience, is compatible with either view. So is the fact that such structures may develop universally. Children the world over may develop the same understandings because they are specified innately, or they may converge on the same conceptions because the crucial evidence is universally the same, and so are theory-formation capacities. All people do, in fact, have minds, and so it is not too surprising that all normal children reach this conclusion about them. Scientists similarly converge on a common account of the world, not because evolutionary theory or Newtonian physics are innately given, but because they are presented with similar patterns of evidence.

On the other hand, the fact that there is some logic in the relations between input and representations is not itself enough to rule out modularity. Evolution

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could seize on these relations also, precisely because they were, at least roughly, the correct ones. Arguments from function are equally undiscriminating. If a particular structure is functionally important from the perspective of the mental life of the individual child, or the cultural life of a community, then it is also likely to be important from an evolutionary perspective and vice-versa. Evolutionary just-so stories and ontogenetic ones are largely interchangeable. This is particularly true when there is a long history of cultural transmission of information.

In short, by considering only a given representational system it may be difficult, if not impossible, to distinguish between these views. The static features of theories, their abstractness, coherence, and explanatory and interpretive force, can also be found in modules. The crucial evidence in this case will be developmental. It is the dynamic features of theories, particularly their defeasibility, that allow us to discriminate them from modules.

In fact, modular accounts of the acquisition of representational systems, are, in an important sense, antidevelopmental (see, for example, Pinker's discussion of "the continuity assumption": Pinker, 1984). Apparent changes in representation occurring over time, on these views, can be accounted for only by processes outside the representational system itself. For example, they may reflect the maturation of another representational system, a later model coming on line. Thus, in his latest position Leslie (1991) acknowledges two transitions in children's representations of the mind. First, there is activation of an initial "metarepresentational" mental module at about 18 months. It is the triggering of this module that stimulates engagement in pretense and an understanding of desires and perceptions at about this age. Then there is the activation of a second, additional theory of mind module at about 3 or 4, which Leslie now postulates to account for children's understanding of belief and reality-appearance at this later age. Similarly, in modular linguistic theories it is sometimes proposed that early language, up until about 3 or 4 when complex syntax appears, is not really language at all (Chomsky, 1980). It instead reflects the operation of a quite different representational system that is supplanted by the maturation of a "real" language aquisition device at about 3.

Alternatively, modularity accounts may invoke external "information-processing limitations" or "performance constraints" to explain development. Fodor (1992) and Leslie (this volume) take this tack. On this view early conceptual structures are just like later ones, but various superficial performance problems prevent children from demonstrating this competence. This is also a common move in linguistic accounts (Chomsky, 1980; Pinker, 1983).

The essence of our preference for a theory theory over an innate module account concerns the developmental data. As described earlier, we believe that the developmental data chart a succession of conceptions of mind, each logically related to earlier conceptions and revealing several intermediate transitional and partial conceptions. On the theory theory the principal

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mechanism for development comes from within the representational system itself; the theory changes as a result of the accumulation of new evidence and also as a result of internal reworkings of the theory itself. According to the theory theory therefore, we would expect to see, not a single privileged representation of the input, but rather a series of developmentally related representations. Erroneous representations that are later restructured should be the rule rather than the exception, under this account. Moreover, as each set of representations is replaced by the next we would expect to see intermediate theory-formation stages and processes. For example, we might first see resistance to and reinterpretation of counterevidence, the invocation of auxiliary hypotheses, and the limited application of ideas in the new theory, before the new theory became completely dominant. We have suggested that just such a sequence is detectable in the child's developing theory of mind.

Such a sequence of several dependent, successor theories, each erroneous in its own way, with characteristic transitions between them, is difficult to explain, in any principled way, by an innate modular maturational account. A postulated succession of two modules, one coming on-line later than and replacing the first seems decidedly ad hoc, in comparison to the theory theory account. Similarly, there is no motivated and general performance limitation account that can describe the pattern of developmental change across the many tasks described above (see Wellman, 1990; Gopnik, 1993). Again the Bartsch and Wellman "discrepant belief" task (see above) is a particularly good indicator of this. The information-processing demands of this task are identical to those of the "false-belief" task, and the task uses precisely the same question. Yet 3-year-olds are consistently quite competent at solving this task. Moreover, erroneous (as opposed to incomplete) representations, which are later modified and restructured, are also difficult to explain on a purely modular account. Evolution might, of course, select for an erroneous representation: The representation just has to be good enough to allow the organism to survive and reproduce. But, if the representational system is good enough, why, on a modularity account, would it be replaced later in development?

It is, of course, logically possible that a maturational sequence of successive modules might just by accident parallel a theory-formation process, and that the triggering inputs just happen to bear the same relation to the privileged representations that evidence bears to theory. Such a view, however, seems unmotivated. It is easy to see how evolution might have selected for an (approximately) correct innately determined representation of the world. It is much more difficult to see how evolution would have selected for a series of representational systems, each maturing separately only to be replaced by another.

What seems more plausible is that evolution selected for a cognitive capacity to revise concepts on the basis of evidence, that is, a theory-making ability. It is also true, as we have said earlier, that an initial state theory, one that influences the later course of development, could be available. Finally, general

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maturational changes in information-processing capacities could enable children to develop some theories rather than others (in much the same way that computers or telescopes enable scientists to develop some types of new theories). Innate capacities could play a role in the development of a theory of mind in all of these ways, but they would not constitute a theory of mind module.

In sum, the role of evidence in theory change, the actively erroneous character of early theories, and the progressive character of theory formation distinguish theory change from the maturation of an innate module. They are also characteristic of children's developing conceptions of mind.

One important sort of evidence that would help differentiate between the theory theory and an indefeasible module is not yet available: intervention studies that would test whether children arrive at different theories if given different patterns of evidence. Experimental studies of this sort are very difficult to conduct. However, examination of the effects of natural variations in evidence are possible, for example, in cross-cultural investigations or investigations of individual and family differences. In the few studies conducted to date young children's performance on appearance-reality (Gardner, Harris, Ohomoto, & Hamazaki, 1988; Flavell et al., 1983) and false-belief tasks (Avis & Harris, 1991) seems quite constant across cultures. However, we do not know the extent of cross-cultural variation in adult theories of mind. If adults converge on the same theory in different cultures, then we might not expect cross-cultural differences in development. Anthropological reports of cross-cultural variation typically concern rather abstract and sophisticated notions of emotional or moral states (Lutz, 1988) or conceptions of self and personhood (Markus & Kitayama, 1991). It is not clear whether cultures do or do not differ much in the sort of very basic psychological knowledge investigated in theory of mind tasks with 3- and 4-year-olds.

Conversely, several suggestive, but still preliminary, studies point to a significant influence of different patterns of evidential input and variation in the timing of the development of mental concepts. Dunn, Brown, Slomkowski, Tesla, & Youngblade (1991) found that children whose parents talked a great deal about mental states developed false-belief explanations earlier than those whose parents did not. Perner, Ruffman, and Leekam (1992) found that children from large families, and particularly younger siblings, did better on false-belief tasks than children with fewer siblings. This is in spite of the fact that older siblings and children from small families do better on general measures of cognitive ability. Perner and his colleagues suggest that siblings may be a particularly relevant source of evidence for the existence of conflicting states of mind.

Theories and domains

Considerable debate in developmental psychology has centered on the extent to which the mind develops in a general versus specific fashion.

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The theory theory provides some important resources for tackling this issue. Again, certain considerations about the nature of theories, articulated by philosophers of science, provide a fruitful starting point for considering naive theories. Of special import here, it seems to us, is a consideration of framework theories in contrast to specific theories (Wellman, 1990). Framework theories, called paradigms, research programs, and research traditions, by philosophers of science, define domains of inquiry. They do so in part by providing an ontology of the fundamental phenomena to be explained: for example, behavior, external stimuli, and overt responses for Behaviorism versus mental states, processes, and intentional action for Mentalism. Specific theories (say the Rescorla–Wagner theory of classical conditioning within Behaviorism) account for specific phenomena by "reducing" them to the ontology and explanatory structures of the framework theory. Notably, framework theories provide a characteristically developmental analysis of theoretical endeavors. A framework theory outlines a domain for discovery – a research program for the development of more detailed explanations and proposals.

This sort of analysis represents an attempt by philosophers of science to characterize an intriguing epistemological structure: framework understandings that predate and inspire, but at the same time are themselves further articulated and developed in more specific understandings of a domain of phenomena. Such structures are both general – encompassing a large domain of phenomena – and specific – specifically different from framework theories for other complementary domains. They are both open-ended and constraining, initiating the development of more specific theories but restricting them by defining the domain of inquiry in the first place.

The theory theory, therefore, predicts a particular combination of specificity and generality. It provides a middle ground between domain-general "stage theories" such as Piaget's on the one hand and accounts of development in terms of skills, or scripts, or "contextual activities" on the other. These latter accounts propose that children develop only an array of different competencies in areas that are highly practiced or repeatedly experienced, an analogy to skill at chess, or understanding restaurants, or sociocultural tasks like weaving. Similarly, some modularity theories propose that there are a great many task- and domain-specific representational systems with relatively little interaction between them (see Cosmides & Tooby, this volume).

A theory, in contrast, and especially a framework theory, should be able to make predictions and explanations across a wide variety of evidence, including evidence that played no role in the theory's initial construction. This predictive and explanatory power is what gives theories their epistemological potency and differentiates theories from skills, rough empirical generalizations, and scripts. Theories typically apply not only to highly familiar or practiced pieces of knowledge but also to brand new ones, sometimes pieces of knowledge that could not have been conceived of or discovered outside of the theory. Equally, we would expect, and indeed have seen, that a theory change

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may affect knowledge of many different sorts in similar ways and at similar times, provided that the theory gives those events a unifying characterization.

Yet theories are limited in their explanatory and predictive scope; even framework theories explain only some kinds of things and not others. Regardless of the extent of the domain, a theory of mind explains only "mental things"; it does not explain physical events and entities – those are left to a theory of physics. Thus there will be several domains in the child's cognition and experience, domains defined by the framework theory that covers them. Wellman and Gelman (1992) suggest, for example, that in the period between 2 and 5 the child's naive psychology is complemented by a naive mechanics and perhaps also a naive biology. These domains are also defined by framework theories; each theory explains only some things and not others and the "things" to be explained are defined by the theory. The entities and laws of these naive theories (objects and physical movements, for example) are very different from those of a theory of mind. The theory theory predicts, therefore, contra stage theories, that individual children's progress in one domain will be quite independent from their progress in the others. Domains outside the scope of some target theory will remain unaffected by changes to that theory, just as we would not expect a new theory of atomic structure to change our theory of evolution or vice versa.

In this way theories not only reflect domains but also actually constitute them. The much-troubled question of how to specify domains is easily solved on this view: A domain is what your framework theory tells you it is. The question is an empirical one and can be addressed by empirically investigating the scope of a person's fundamental theories, such as the individual's theory of mind.

An important consequence of this view is that domains themselves may change as theories change. Once again this differentiates theories from modules. For example, the Berkeley biologists recently reorganized themselves, at great expense, into a different set of departments. Zoology, botany, physiology, and paleontology were replaced by cell biology, biochemistry, immunology, and genetics. Scientific advances in biology meant that apparently natural domains (plants vs. animals, for instance) had been subsumed by others defined by different theoretical considerations. The new departments reflect a change in the framework theories within biology as a whole. Similarly, some of the changes we have sketched in this chapter seem to us to involve changes in children's very idea of the scope and nature of psychological explanation. The domain at issue is changing with development.

A theory approach to domain specificity provides a cognitive analysis of domains rather than a neurological-biological one. Ultimately, of course, cognition is a product of neurology. And there is evidence to suggest the existence of certain neurologically specifiable domains of thought (see other chapters in this volume). However, it remains to be seen how various psychological domains relate to neurologically specified domains, and it seems unlikely

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to us that all interesting cognitive domains will be the result of distinct neurological structures. For example, connectionist models demonstrate the possibility of single neural structures that can perform several separate psychological tasks. In general, we do not think that cognitive domains will neatly reduce to neurological domains, just as psychology will not neatly reduce to neurology. We need cognitive analyses to identify cognitive domains. The theory theory offers tools for this task.

Conclusion

In spite of Piaget's arguments for a "genetic" epistemology, many recent approaches to cognitive development, including those information-processing approaches misleadingly called neo-Piagetian, are antiepistemological in a deep sense. That is, they explain behavioral changes not by describing the endogenous development of knowledge itself but by invoking exogenous mechanisms, including maturation, triggered changes in information-processing skills, socially provided knowledge, or tools. Such mechanisms must indeed be responsible for various developmental changes. But the changes we are interested in cannot be accounted for in these ways alone, because they involve changes in the structure of knowledge itself and not simply in its instantiation or application. The philosopher of science's notion of a theory and characterization of theory change provide crucial conceptual tools for talking about such deep cognitive changes, and, as we have tried to show here, such notions provide illuminating insights into children's actual behavior. It is of course true that within philosophy and history of science, theory formation and theory change are far from well understood. In important ways the scientist's ability to learn about the world is still almost as mysterious as the child's. Nevertheless, it seems to us that reducing two mysteries to one is an important advance, and a great deal more than we usually achieve.

Notes

1. Having a theory also need not mean awareness that what you have is a theory. Neither children nor scientists may have much conscious access to the epistemological processes in which they engage. In fact, philosophers of science often dismiss scientists' accounts of their own theory constructing activities as incorrect. By and large, scientists' theories are more explicitly held than children's but even this may be more a matter of degree than kind. In explanation studies, and in natural conversation, 3- and 4-year-olds are quite explicit about their convictions about beliefs and desires and the relations between them. Conversely, it is often true that historians and philosophers of science "reconstruct" theories from the actual writings and practice of science. Abstracting Kepler's heliocentric theory, for example, from the welter of mysticism, superstition, and just plain confusion in his writing may not be all that different than abstracting the 3-year-old's theory of belief from her language and behavior. What children or scientists say about

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what they know and do is a useful source of information but in neither case is it definitive.

2. It is important to be as clear as possible about the way in which theoretical constructs are abstract, unobservable, and postulated. We mean abstract in the sense of "thought of apart from" observable particularities – we do not mean abstruse or merely ideal. Theoretical entities are unobservable in the sense of not being obviously a part of the evidential phenomena to-be-explained, not that they are necessarily incapable of being observed in any fashion whatsoever. Thus, we could postulate that genes control inherited features such as eye color and height, in order to provide a theoretical account, and still fully expect that genes are observable in some fashion. It is simply that genes are not directly evident in, observable in, the phenomena of eye color and height themselves. Similarly, postulated does not mean conjured out of thin air; it means recruited for explanatory purposes from outside the evidential phenomena themselves. Thus (natural) selection can be postulated to account for the origin of species but at the same time selection can be fully concrete and observable, in the realm of animal breeding, for example. It is the recruitment of selection to account for natural speciation that is postulational; selection itself is not a mere postulated entity.
3. On the theory theory, the ages of development are not crucial. In fact we would expect to find, as indeed we do, wide variation in the ages at which successive theories develop. We would expect to find similar sequences of development, however. We will use ages as a rough way of referring to successive theories.
4. In some modular systems, there are several alternative branching routes, so to speak, that determine the eventual form the module may take. These are generally described as "parameters" set by the input. The relation between input and the setting of the parameter is still, however, a relation of triggering. In contrast, in a theory theory, by analogy with scientific theories, there should be very wide scope for novel theories, not simply a choice of several options.

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